

Base and azimuth drive: virtual fit-up

The assembled view shows the saved component positions. The exploded view separates the pieces to show their order; its offsets are illustrative.

BASE AND AZIMUTH DRIVE · ASSEMBLY 1

Eleven named parts, including four provisional envelopes/blanks. Exploded offsets show order only.

ASSEMBLED POSITION



EXPLODED ORDER



Fixed support: 01 → 02 → 03 • Rotating stack: 04 → 31 → 05 → 06 → 07 • Drive: 26 → 27 → 28

All dimensions are provisional millimetres. Bearing, motor, gear teeth and torque connection require selected hardware.

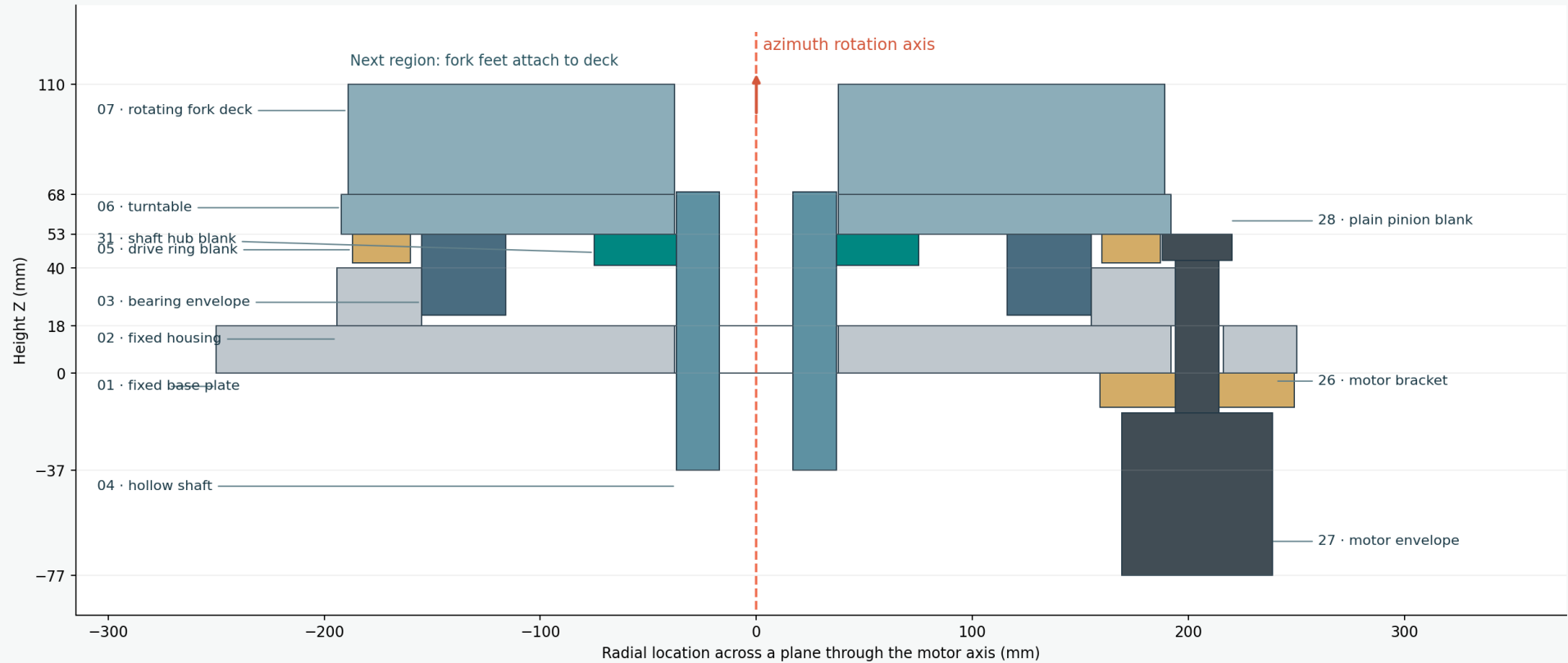
Source: image-derived concept model. Eleven named STEP parts. Dimensions are provisional millimetres.

How the parts stack on the azimuth axis

This section passes through the azimuth axis and the motor location. Its colours and geometry explain the interfaces; it is not a manufacturing drawing.

BASE SUBASSEMBLY · SCHEMATIC AXIAL SECTION

Part numbers map to the STEP files. Colours identify interfaces; the section is simplified for explanation.



The bearing and motor are envelopes. Gear teeth, fasteners, bearing preload and shaft-to-hub torque connection are not designed.

Read from bottom to top: fixed plate 01, bearing housing 02 and bearing envelope 03, rotating hub 31 and drive ring 05, turntable 06, then fork deck 07. The hollow shaft 04 passes through the centre.

Parts and suggested assembly order

The new Base_Subassembly.step contains eleven separately named parts from the full gimbal. It starts with the fixed circular base and ends at the rotating rectangular deck. Base_Exploded_View.png shows the order schematically. This is the second, interface-aligned concept revision. Replace the earlier 30-part package with Gimbal_CAD_v2_With_Base_Assembly.zip when working on the base.

Coordinates: millimetres. Z is the vertical azimuth axis. The underside of the fixed base plate is $Z = 0$. X runs along the future altitude axis; Y is front/back. The azimuth motor is at $X = 120$, $Y = -165$ mm.

Parts included

Stage	Part numbers	Purpose
Fixed support	01, 02, 03	Base plate, housing and provisional bearing annulus
Rotating stack	04, 31, 05, 06, 07	Hollow shaft, separate hub, drive ring, turntable and fork deck
Drive	26, 27, 28	Azimuth motor bracket, motor envelope and plain pinion blank

Suggested assembly order

- 1. Orient and anchor part 01. The $\varnothing 500 \times 18$ base plate has twelve $\varnothing 9$ foundation holes on an assumed $\varnothing 460$ pitch circle. Its central $\varnothing 76$ passage provides clearance for the vertical shaft envelope. Confirm the actual foundation pattern before drilling anything.
- 2. Fit part 02 to part 01. Eight $\varnothing 10$ holes on an assumed $\varnothing 350$ pitch circle now align between the azimuth bearing housing and the fixed plate. The housing spans $Z = 18\text{--}40$ mm. Fastener type, seating face and torque remain to be specified.
- 3. Place part 03 in the housing. Its $\varnothing 310$ outer envelope meets the housing's nominal $\varnothing 310$ seat and extends to the underside of the turntable at $Z = 53$ mm. This is a *space claim*, not a selected rolling-element bearing. The actual bearing determines fit, retention, preload and life.
- 4. Prepare the rotating stack. Part 04 is the hollow vertical shaft. Part 31 is a separate hub blank below the turntable ($Z = 41\text{--}53$). Its six $\varnothing 8$ holes on $\varnothing 120$ PCD match part 06. The shaft-to-hub key, clamp or other torque connection is not yet defined.
- 5. Join the drive ring and turntable. Part 05 meets the underside of part 06 at $Z = 53$ mm. Their eight $\varnothing 10$ holes on $\varnothing 348$ PCD align. The ring is a smooth blank; the eventual gear/tooth design may alter its radius and attachment.
- 6. Seat the rotating stack on the bearing. Part 06 meets the upper bearing envelope at $Z = 53$ mm. Check the vertical axis, the chosen bearing's axial/radial retention and clearance, and the moving parts' free rotation after actual bearing data are available.

- 7. Attach the deck. Part 07 sits on part 06 at $Z = 68$ mm. Four $\varnothing 9$ holes at $X = \pm 130$, $Y = \pm 90$ mm align between these parts. Its top is $Z = 110$ mm; the next region will attach the fork feet at $X = \pm 154$, $Y = \pm 55$ mm.
- 8. Add the underside motor group. Part 26 meets the underside of the fixed plate. Its four $\varnothing 9$ holes align with four fixed-plate holes. Part 27 occupies the motor volume; part 28 passes through the $\varnothing 24$ drive-shaft opening. These are envelopes pending an actual gearbox and shaft interface.
- 9. Check motion and records. In the CAD model, inspect radial clearance between the fixed bearing housing and the rotating drive ring, check that hole patterns remain aligned, and record the bearing, gear, fastener and motor choices before physical assembly.

What the fit-up establishes

- The housing-to-base, ring-to-turntable, hub-to-turntable, turntable-to-deck and motor-bracket-to-base hole centers are now matched in the exported solids.
- The bearing annulus reaches the turntable underside, and the nominal bearing/housing diameters meet.
- The ring and pinion are untoothed blanks with a deliberate gap; gear mesh has not been designed or validated.
- Individual STEP parts and Base_Subassembly.step were re-imported and checked as valid solids/assembly entities.

Dimensions required before a physical build

Select the azimuth bearing and supply its OD, ID, width, loads and mounting instructions; the azimuth motor/gearbox and output shaft; the drive ratio and backlash requirement; the mass and center of gravity of the complete rotating assembly; foundation bolt pattern; materials, joining method, environmental and pointing requirements. These determine fits, fastener lengths and grades, preload, torque, hub/key design, stiffness and safety margins. None can be set reliably from the reference image alone.